DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

T. B. C. : GPGM – 5/20

Test Booklet Series

A

TEST BOOKLET

MINING OFFICER

(APPLIED GEOLOGY)

PAPER – II

Sl. No. 50009

Time Allowed : 3 Hours

Maximum Marks : 300

: INSTRUCTIONS TO CANDIDATES :

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET **DOES NOT** HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET OF THE SAME SERIES ISSUED TO YOU.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES **A, B, C** OR **D**, AS THE CASE MAY BE, IN THE APPROPRIATE PLACE IN THE ANSWER SHEET USING BALL POINT PEN (BLUE OR BLACK).
3. You have to enter your **Roll No.** on the Test Booklet in the Box provided alongside. **DO NOT** write *anything else* on the Test Booklet.
4. **YOU ARE REQUIRED TO FILL UP & DARKEN ROLL NO., TEST BOOKLET / QUESTION BOOKLET SERIES IN THE ANSWER SHEET AS WELL AS FILL UP TEST BOOKLET / QUESTION BOOKLET SERIES AND SERIAL NO. AND ANSWER SHEET SERIAL NO. IN THE ATTENDANCE SHEET CAREFULLY. WRONGLY FILLED UP ANSWER SHEETS ARE LIABLE FOR REJECTION AT THE RISK OF THE CANDIDATE.**
5. This Test Booklet contains **150** items (questions). Each item (question) comprises four responses (answers). You have to select the correct response (answer) which you want to mark (darken) on the Answer Sheet. In case, you feel that there is more than one correct response (answer), you should mark (darken) the response (answer) which you consider the best. In any case, choose **ONLY ONE** response (answer) for each item (question).
6. You have to mark (darken) all your responses (answers) **ONLY** on the **separate Answer Sheet** provided by **using BALL POINT PEN (BLUE OR BLACK)**. See instructions in the Answer Sheet.
7. All items (questions) carry equal marks. All items (questions) are compulsory. Your total marks will depend only on the number of correct responses (answers) marked by you in the Answer Sheet. **There will be no negative marking for wrong response (answer).**
8. Before you proceed to mark (darken) in the Answer Sheet the responses (answers) to various items (questions) in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per the instructions sent to you with your **Admission Certificate**.
9. After you have completed filling in all your responses (answers) on the Answer Sheet and after conclusion of the examination, you should hand over to the Invigilator the *Answer Sheet* issued to you. You are allowed to take with you the candidate's copy / second page of the Answer Sheet along with the **Test Booklet**, after completion of the examination, for your reference.
10. Sheets for rough work are appended in the Test Booklet at the end.

SEAL

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

1. Tornadoes are common in which of the following countries ?

   (A) India

   (B) Bangladesh

   (C) China

   (D) U. S. A.

2. Landslides often occur in :

   (A) Desert region

   (B) Forest region

   (C) Tundra region

   (D) Hilly region

3. Which of the following is not a natural disaster ?

   (A) Nuclear explosion

   (B) Deforestation

   (C) Forest fire

   (D) Lighting

4. The word Tsunami has been derived from :

   (A) Sanskrit

   (B) French

   (C) Chinese

   (D) Japanese

5. All of the following are TRUE about disasters EXCEPT :

   (A) A disaster may be domestic or international

   (B) A disaster may be caused by nature or have human origins

   (C) A disaster always receives widespread media coverage

   (D) A disaster may have a known and gradual onset

6. Disasters frequently result in all of the following EXCEPT :

   (A) Damage to the ecological environment

   (B) Displacement of populations

   (C) Destruction of a population's homeland

   (D) Sustained public attention during the recovery phase

7. Social workers skilled in crisis management work on :

   (A) Tornado or flood

   (B) Violent events such as child abuse, domestic abuse, crime

   (C) Psychopathology that triggers a crisis i.e. suicide attempt, drug overdose

   (D) Car accident, life-threatening illness

8. According to the United Nations beween 1992 and 2000, how many disasters (approximately) occurred per year throughout the world ?

(A) 100

(B) 300

(C) 500

(D) 800

9. According to Reza et al, the number of people who have lost their lives in the 25 largest violent events of the 20th century number :

(A) Almost 1 million

(B) Almost 10 million

(C) Almost 100 million

(D) Almost 200 million

10. Which one of the following theories of disaster management informs organizational readiness and response in a disaster ?

(A) Hobfoll's theory of Conservation of Resources

(B) The problem-solving "task model"

(C) Structure-functional theory

(D) Theory of traumatogenic forces

11. Energy is released from fossil fuels when they are :

(A) Pumped

(B) Cooled

(C) Burned

(D) Pressurized

12. The most nuclear fuel used in the world is :

(A) Thorium – 232

(B) Uranium – 238

(C) Uranium – 235

(D) Plutonium – 239

13. In the production of wave energy, which form of energy is used ?

(A) Potential energy

(B) Kinetic energy

(C) Solar energy

(D) Wind energy

14. Which of the following is a non-renewable energy resource ?

(A) Solar

(B) Methane

(C) Hydro electric

(D) Coal

15. A tidal barrage is a barrier built over a :

(A) River bed

(B) River estuary

(C) River end

(D) River starting

16. In hydroelectricity power which of the following statement is true ?

(A) Kinetic energy is transferred to potential

(B) Potential energy is transferred to kinetic

(C) Solar energy is transferred to wind energy

(D) Wind energy is transferred to solar energy

17. In order to produce solar energy during sunlight, where the energy is stored in the batteries ?

(A) Nickel sulphur

(B) Zinc Cadmium

(C) Nickel Cadmium

(D) Nickel Zinc

18. How many forms of fossil fuels exists ?

(A) One

(B) Two

(C) Three

(D) Four

19. According to WHO, how many premature deaths annually linked to air pollution causing by the burning of fossil fuels ?

(A) One milliion

(B) Three milliion

(C) Five milliion

(D) Seven milliion

20. Energy in the form of heat and light is obtained by :

(A) Biomass

(B) Fossil fuels

(C) Sun

(D) Wind

21. SI unit for energy is :

(A) Watt

(B) Kilogram

(C) Newton

(D) Joule

22. Trapped heat inside the earth is known as :

(A) Heat energy

(B) Kinetic energy

(C) Geothermal energy

(D) Thermal energy

23. The efficiency of a nuclear power plant in comparison to a conventional thermal power plant is :

(A) Same

(B) More

(C) Less

(D) May be less or more depending on size

24. An assemblage of high temperature metamorphic gangue minerals in contact metasomatic deposits is known as :

(A) Skarn

(B) Gondite

(C) Gossans

(D) Bar deposits

25. The branch of Geology that deals principally with the mode of occurrence, origin, distribution, classification and association of minerals and rocks of economic importance is known as :

(A) Engineering geology

(B) Structural geology

(C) Mineralogy

(D) Economic geology

26. A deposit must contain at least certain percentage of metal in it and lowest admissible limit of metallic content of an ore known as :

(A) Simple ore

(B) Ore mineral

(C) Tenore

(D) Economic ore

27. In metalliferous deposit, the ore minerals generally occur in association with some non-metallic minerals is known as :

(A) Gangue

(B) Cap rock

(C) Ore shoots

(D) Late magmatic mineral deposit

28. The mineralized vertical solution joints in carbonate country rock containing valuable deposits of metallic and non-metallic minerals is known as :

(A) Ladder vein

(B) Saddle reefs

(C) Gash vein

(D) Grouting

29. The different geological periods during which mineral deposits were formed in abundance in certain sectors of globe are commonly known as :

(A) Syngenetic period

(B) Metallogenetic period

(C) Epigenetic period

(D) Sublimation period

30. The mineral deposit in solid rock is known as :

(A) Ore

(B) Lode

(C) Gangue

(D) Lithology

31. A natural mineral substance of iron oxide, clay and sometimes manganese is known as :

(A) Shale

(B) Silt

(C) Ochre

(D) Tuff

32. Iron ore deposits of Goa are mainly of :

(A) Magnetite type

(B) Hematite type

(C) Blue dust

(D) Both magnetite and hematite types

33. Which of the following are most useful varieties of the asbestos ?

(A) Chrysolite, Amosite and Crocidolite

(B) Chrysolite and Acnnome

(C) Amosite, Actinolite and Tremolite

(D) Actinolite, Anthophyllite and Tremolite

34. One of the plates of the apical disc in an echinoid shell is perforated, this plate is known as :

(A) Ocular plates

(B) Genital plate

(C) Med reproach plate

(D) Ambulacra plate

35. Stephen J. Gould is a palaeontologist who speculated that, of all the organisms that ever lived on earth only, how many % are still alive ?

(A) 1

(B) 20

(C) 50

(D) 99

36. In palaeontology, the conversion of remains of plants and animals into rock is known as :

(A) Fossils

(B) Petrification

(C) Moulds

(D) Carbonization

37. An animal is more likely to become fossilized if :

(A) The animal dies at sea and sinks deep

(B) Left exposed to the elements

(C) The animal is a soft-bodied animal like jelly fish

(D) The animal is hard-bodied

38. Which of the following statements is true ?

(A) Ground water percolation can hinder mineralization of bone

(B) Bones usually contain organic as well as inorganic molecules

(C) Jelly fish can become fossils as their body contains hard parts

(D) Bones contains only organic molecules

39. Which of the following cannot be inferred from studying the fossilized skeletons of animals ?

(A) Pathologies

(B) Life expectancy

(C) Growth pattern

(D) Decay rate

40. Palaeoanthropology is the study of :

(A) Fossils of early birds and their ancestors

(B) Fossils of early humans and their ancestors

(C) Fossils of early fish and its descendants

(D) Fossils of early reptiles and its descendants

41. What is a holotype ?

(A) A single physical example of an organism known to have been used when the species was described

(B) A term used to describe special type of bones found exclusively in birds

(C) A recently formed fossil specimen

(D) A long-ago forms fossil pattern

42. Radiocarbon dating can help find the age range of biological specimens no older than :

(A) 1,00,000 years

(B) 50,000 years

(C) 5,00,000 years

(D) 10,00,000 years

43. The scientific study of the structure of bones, skeletal elements and microbone morphology is known as :

(A) Osteology

(B) Herpetology

(C) Entomology

(D) Micrology

44. It is an important constituent of granite and a chief constituent of sandstone where it occurs as broken form. Its hardness is 7, name the mineral :

(A) Magnetite

(B) Topaz

(C) Orthoclase

(D) Quartz

45. Which of the following is a buff-coloured deposit made up of angular fragments of unaltered mineral grains and is characteristically unstratified, the thickness of such deposit may vary from few inches to hundreds of feet ?

(A) Runoff

(B) Flyoff

(C) Pot holes

(D) Loess

46. The ratio of volume of voids in a rock or soil to its total volume is known as :

(A) Lithology

(B) Plasticity

(C) Permeability

(D) Porosity

47. Which of the following is not a mineral ?

(A) Olivine

(B) Calcite

(C) Bauxite

(D) Quartz

48. The hardness of Corundum is :

(A) 3

(B) 8

(C) 9

(D) 1

49. The quartz mineral belongs to :

(A) Silicate

(B) Oxide

(C) Carbonate

(D) None of these

50. The streak of haematite is :

(A) Grey

(B) Black

(C) Cherry red

(D) Red

51. Admantine lustre is shown by :

(A) Calcite

(B) Felspar

(C) Sphalerite

(D) Diamond

52. Crystallography is a branch of :

(A) Geology

(B) Minerology

(C) Physical geology

(D) Both (A) and (B)

53. Which mineral is an iron ore ?

(A) Bauxite

(B) Malachite

(C) Haematite

(D) Galena

54. It is more or less parallel sided vertical wall of igneous rock formed due to upward intrusion of molten magma through fissures or cracks and is a common occurrence in coalifields. What it is known as ?

(A) Sill

(B) Dyke

(C) Lava

(D) Laccolith

55. The plane of contact between the two series of strata is defined as :

(A) Unconfined

(B) Heterogeneity

(C) Homogeneity

(D) Dip and strike

56. In respect of fault, the vertical displacement of the fractured bed is known as :

(A) Heave

(B) Hade

(C) Fault plane

(D) Throw of fault

57. A fault that runs at some angle between the dip and strike of the bed is known as :

(A) Dip fault

(B) Oblique fault

(C) Strike fault

(D) Throw of fault

58. The wavy undulations developed in country rocks whenever the region is subjected to severe pressure or stress are known as :

(A) Fault

(B) Fold

(C) Dyke

(D) Sill

59. When axial plane of a fold is vertical and either of the limbs have the same amount of dip then such fold is described as :

(A) Asymmetric

(B) Fan

(C) Symmetric

(D) Chevron

60. The horizontal component of the apparent displacement of a bed, measured along the direction of dip of the fault is known as :

(A) Throw of fault

(B) Heave of fault

(C) Hade of fault

(D) Horsts

61. A part of any stream or rock that is exposed at the surface is known as :

(A) Fault

(B) Fold

(C) Outcrop

(D) Dyke

62. In a dip-slip fault, if the hanging wall block moved up relative to the footwall block, then the fault is classified as :

(A) Normal fault

(B) Reverse fault

(C) Steps fault

(D) Downward faults

63. The crest of fold where strata are bent up to form an arch is known as :

(A) Syncline

(B) Swilley

(C) Outlier

(D) Anticline

64. How thick is the crust of the Earth ?

(A) About 4 km

(B) About 4 miles

(C) About 40 kms

(D) About 400 kms

65. Which layer separates crust from core ?

(A) Magma layer

(B) Mantle

(C) Lithosphere

(D) Continent

66. What drives the Earth's internal heat engine ?

(A) Radioactivity

(B) Solar energy

(C) Volcanoes

(D) Ocean tides

67. Which of the following compound is most abundant in earth crust ?

(A) Silicon oxide

(B) Hematite

(C) Magnetite

(D) Cupric oxide

68. Which is the last mineral to crystallize from a magma in Bowen's reaction series ?

(A) Olivine

(B) Quartz

(C) Pyroxene

(D) Biotite

69. What is the volcanic equivalent of the gabbro ?

(A) Rhyolite

(B) Gabbro

(C) Andesite

(D) Basalt

70. Who is the father of modern geochemistry ?

(A) Diels-Alder

(B) Ludwig gatter man

(C) Fidel craft

(D) Victor Moritz gold schmidt

71. Name the chemistry that deals with geological process at the "atomic level" and the history of atoms in the earth's crust and on the planet as a whole :

(A) Organic chemistry

(B) Geochemistry

(C) Physical chemistry

(D) Inorganic chemistry

72. The composition, structure, process and other physical aspects of Earth are studied by :

(A) Geochemists

(B) Pharmacologists

(C) Forensic scientists

(D) Analytical chemists

73. What is the percentage of composition of SiO_2 on earth's crust ?

(A) 55.71

(B) 57.71

(C) 59.71

(D) 63.71

74. How many elements are naturally occurring in earth's crust ?

(A) 91

(B) 92

(C) 93

(D) 94

75. The most nuclear fuel used in the world is :

(A) Thorium – 232

(B) Uranium – 238

(C) Uranium – 235

(D) Plutonium – 239

76. In metalliferous deposit, the ore minerals generally occur in association with some none metallic minerals is known as :

(A) Late magmatic mineral deposit

(B) Cap rock

(C) Ore shoots

(D) Gangue

77. The branch of geology which deals with descriptions, classification and correlations of strata in sedimentary rock is known as :

(A) Stratigraphy

(B) Sedimentary petrology

(C) Physical geology

(D) Igneous petrology

78. The hollows left within the rock beds are filled up with mineral matter giving rise to :

(A) Moulds

(B) Casts

(C) Fossils

(D) Imprints

79. To theories are known to explain the heavy accumulation of vegetable and organic matters that formed coal seams. If one of the theories is draft theory, what is the other ?

(A) Beam theory

(B) Keystone theory

(C) In-situ theory

(D) Stratified theory

80. Dull, fribal appearance that soil the finger and most impure band in coal is known as :

(A) Clarain

(B) Fusain

(C) Durain

(D) Vitrain

81. As per the size gradation of coal, the size of rubble coal is :

(A) 51 mm to 200 mm

(B) 25 mm to 50 mm

(C) 0 mm to 24 mm

(D) Any unscreened size

82. In order to calculate useful heat velue in Keal/kg of non-coking coal, beside ash percentage what is the name of other character which is considered ?

(A) Carbon

(B) Moisture

(C) Volatile

(D) Sulphur

83. Coalification has taken place in order of :

(A) Peat, lignite, bituminous and anthracite

(B) Lignite, peat, bituminous and anthracite

(C) Peat, lignite, anthracite and bituminous

(D) Bituminous, peat, lignite and anthracite

84. Coal seams are normal found in :

(A) Sedimentary rocks

(B) Igneous rocks

(C) Metamorphic rocks

(D) Any type of rocks

85. If sandstone is the original rock, then what is its metamorphic rock ?

(A) Slate

(B) Gneiss

(C) Quartzite

(D) Marble

86. The rock which are formed by erosion, weathering transportation and ultimately deposition of pre-existing of rock is known as :

(A) Igneous rock

(B) Sedimentary rocks

(C) Metamorphic rocks

(D) Both (A) and (B)

87. The mineral deposit in soiled rock is called :

(A) Ore

(B) Lode

(C) Gangue

(D) Lithology

88. During which transportation, the organism is decomposed and it loses nitrogen, oxygen and other volatile constituent ?

(A) Evolution

(B) Sedimentation

(C) Crystallization

(D) Carbonization

89. It is a compound of $CaCO_3$ and $MgCO_3$ in various proportion :

(A) Magnetite

(B) Laterite

(C) Siderite

(D) Dolomite

90. Rocks that are formed by the cooling of molten material (called magma) at or relatively near the earth surface is called :

(A) Igneous rocks

(B) Sedimentary rocks

(C) Metamorphic rocks

(D) Crystalline

91. On the basis of which content, the igneous rocks are classified as acid rocks and basic rocks ?

(A) Magnetite

(B) Hematite

(C) Bauxite

(D) Silica

92. In metalliferous deposit, the ore minerals generally occur in association with some none metallic minerals is known as :

(A) Late magmatic mineral deposit

(B) Cap rock

(C) Ore shoots

(D) Gangue

93. The mineralized vertical solution joints in carbonate country rocks containing valuable deposits of metallic and non-metallic minerals is known as :

(A) Ladder vein

(B) Saddle reefs

(C) Gash vein

(D) Grouting

94. The sources of mineral deposits are :

(A) Volcanoes

(B) Fumarols

(C) Hot springs

(D) All of these

95. The concentration chromity deposition in ultra-basic rock is an example of :

(A) Dissemination deposit

(B) Early magmatic segregation deposit

(C) Liquid injection deposit

(D) Immiscible liquid deposition

96. The high temperature deposits are :

(A) Mesothermal

(B) Hypothermal

(C) Epithermal

(D) None of these

97. Which mineral is an aluminium ore ?

(A) Bauxite

(B) Melachite

(C) Hematite

(D) Galana

98. Plane strain loading means, normal strain in that direction is :

(A) Zero

(B) Always negative value

(C) Always positive value

(D) Sometimes negative and sometimes positive

99. In Mohr's circle, one axis is shear stress and another axis is :

(A) Tensile stress

(B) Triaxial stress

(C) Normal stress

(D) Compressive stress

100. Porosity of rock sample is defined as ratio of the volume of voids to the :

(A) Volume of solids

(B) Total volume of sample

(C) Volume of water

(D) None of these

101. When unconfined compressive strength of a rock specimen is determined, patterns of cracks developed will be :

(A) Horizontal

(B) Diagonal

(C) Vertical

(D) No specific direction / pattern

102. Which of the following method is used to measure internal stresses inside a rock mass ?

(A) Flat-jack method

(B) Brazilian test

(C) Shear test

(D) Tri-axial test

103. Generally, rock bursts occur in which of the following areas / zone ?

(A) Low stress area / zone

(B) High stress area / zone

(C) No stress area / zone

(D) At shallow depths

104. Poisson ratio refers to :

(A) Rate of change of strain as a function of stress

(B) Load applied on the material per unit area

(C) Ratio of lateral strain by longitudinal strain

(D) Load applied on the material per unit volume

105. Brazilian test to determine 'Tensile strength' of rock samples use core specimen having length: dia. ratio in the range of :

(A) 0.5 to 1.0

(B) 1.2 to 1.5

(C) 1.5 to 2.0

(D) 2.0 to 2.5

106. How many principal stresses a homogenous, linear matrix equation developed in a stress field has ?

(A) 1

(B) 2

(C) 3

(D) 5

107. Which of the following behaviour describe quantitatively the stress-strain rates related in a material under load ?

(A) Constitutive behaviour

(B) Homogeneous behaviour

(C) Complex behaviour

(D) Bieniawski's in-situ behaviour

108. Which of the following diagram can represent one aspect of biaxial stress action on one point ? With respect to set of reference axes :

(A) Mohr's circle diagram

(B) Rose diagram

(C) Infra formal shears diagram

(D) Goodman diagram

109. Deere introduced this concept of quantify discontinuity spacing in rock masses of which of these ?

(A) RMR

(B) RQD

(C) Roof class

(D) JRC

110. Name the diagram where the strikes of discontinuties are oriented at about 5° interval :

(A) Mohr's circle diagram

(B) Coward diagram

(C) Hock and Brown diagram

(D) Rose diagram

111. The plotting of planes and their poles is carried out with the aid of :

(A) Great circle

(B) Lambert

(C) Stereo net

(D) Counting net

112. Which of the following is known as point load index ?

(A) (14 + 0.175D) Is

(B) P/D^2

(C) $\sigma_m = 1/3$

(D) $S = C + \sigma \tan \theta$

113. Which of the following test may be used to provide an indirect estimate of uniaxial compressive strength ?

(A) Point load test

(B) Servo controlled test

(C) Displacement test

(D) Post peak test

114. It is easy to integrate GIS, Remote Sensing and GPS technologies because these are :

(A) Digital, Special and Generic

(B) Digital, Analogue and Manual

(C) Digita, Spatial and Generic

(D) Negative, Positive and Neutral

115. Remote Sensing is unique because it provides :

(A) Synoptic view

(B) Special information

(C) Superior information

(D) Encrypted information

116. Which of the following is an area of Remote Sensing application ?

(A) Dispatch of emergency medical services

(B) Egg Production

(C) Mapping and water resources

(D) Census

117. The purpose of georeferencing is to :

(A) Converting data to a feature class

(B) Projecting your data so that it has no distortion

(C) Assigning geographic location information

(D) Combining two data set

118. A cell with 1m spatial resolution covers an area of $1m^2$ spatial resolution covers an area of :

(A) $0.1\ m^2$

(B) $0.001\ m^2$

(C) $10\ cm^2$

(D) $100\ cm^2$

119. Resampling of a raster image :

(A) Increases the accuracy

(B) Increases the precision

(C) Decreases data details

(D) Increases data details

120. In geographic coordinate system 1° at equator is equivalent to :

(A) 43.49 km

(B) 78.71 km

(C) 111.32 km

(D) 102.47 km

121. The part radiation due to scattered/ diffused radiation entering the field of view of a remote sensor other than that from the required target :

(A) Increases the contrast of the image but reduces the sharpness

(B) Reduces the contrast of the image and also its sharpness

(C) Increases both the contrast and sharpness

(D) Reduces the contrast but increases the sharpness

122. There are two Survey of India 1 : 50,000 toposheets with 15' X 15' angular coverage, one from Jammu and another from Kerala. Which of the following statement is true ?

(A) Both the toposheets will cover exactly same size of area

(B) The toposheet from Jammu will cover a larger area that of the Kerala

(C) The toposheet from Kerala will cover a larger area than that of the Jammu

(D) Projection will affect the size of area coverage

123. GIS, Remote Sensing and GPS technologies are :

(A) Generic, Digital and Spatial

(B) Manual, Spatial and Digital

(C) Analogue, Manual and Spatial

(D) Generic, Analogue and Spatial

124. GPS measurement is accurate to within 25 meters (95% confidence). This means that you can be 95% sure that your measurement is somewhere within a meter error circle and there is a 5% chance that the error is larger than :

(A) 2.5

(B) 5

(C) 25

(D) 50

125. The shape of a pixel of an image can only be :

(A) Rectangular

(B) Circular

(C) Square

(D) Triangle

126. A and B are two towers of equal height diametrically opposite on either side of the nadir point, at 3 km and 5 km distances. Which one of the following statements is correct ?

(A) Height displacement of A will be less than that of B

(B) Height displacement of B will be less than that of A

(C) Height displacement of A and B is equal

(D) Height displacement of A and B will be towards each other

127. Who coined the term, 'Remote Sensing' ?

(A) Evelyn L. Pruitt, a geographer

(B) Gaspard Felix Tourna chon, a French scientist

(C) Wilbur Wright, an Italian scientist

(D) Albert Einstein

128. Which one of the following geometric errors of satellite sensors is random ?

(A) Scan skew

(B) Panoramic distortion

(C) Earth rotation

(D) Altitude variation

129. The altitudinal distance of a geostationary satellite from the earth is about :

(A) 26,000 km

(B) 30,000 km

(C) 36,000 km

(D) 44,000 km

130. Which of the following wave is not employed in case of Remote Sensing ?

(A) X-ray

(B) Thermal IR

(C) Visible ray

(D) Radio waves

131. GIS stands for :

(A) Geographic Information System

(B) Generic Information System

(C) Geological Information System

(D) Geographic Information Sharing

132. GIS deals with which of the following data ?

(A) Numeric data

(B) Binary data

(C) Spatial data

(D) Complex data

133. Which of the following statements is true about the capabilities of GIS ?

(A) Data capture and preparation

(B) Data management, including storage and maintenance

(C) Data manipulation and analysis

(D) Data presentation

134. A (geographic) field is a geographic phenomenon for which, for every point in the study area :

(A) A value can be determined

(B) A value cannot be determined

(C) A value is not relevant

(D) A value is missing

135. GIS uses the information from which of the following sources ?

(A) Non-spatial information system

(B) Spatial information system

(C) Global information system

(D) Position information system

136. Which of the following formats can be used for GIS output ?

(A) DXF

(B) PDF

(C) GIF

(D) HTML

137. Among the following, which do not come under the components of GIS ?

(A) Hardware

(B) Software

(C) Compiler

(D) Data

138. The fundamental principle which refers to the fact that locations that are closer together are more likely to have similar values than locations that are far apart, is commonly referred to as :

(A) Tobler's first law of Geography

(B) Kepler's first law of Geography

(C) Anthony's first law of Geography

(D) Thompson first law of Geography

139. Interpolation is made possible by a principle called :

(A) Spatial Autocorrelation

(B) Spatial auto-correction

(C) Thematic Autocorrelation

(D) Thermatic auto-correction

140. What is 'Metadata' ?

(A) It is 'data about data'

(B) It is 'meteorological data'

(C) It is 'oceanic data'

(D) It is 'contour data'

141. The oil can be trapped in the :

(A) Structure trap only

(B) Structure and stratigraphic traps

(C) Stratigraphic traps only

(D) Movable traps

142. Which of the following is not a stratigraphic oil trap ?

(A) Unconformity

(B) Reef

(C) Anticline

(D) Pinch out

143. Which of the following is not an evaporite rock ?

(A) Gypsum

(B) Evaporite

(C) Muscovite

(D) Salt

144. The movement of the oil from the source rock to the reservoir rock is known as :

(A) Primary migration

(B) Secondary migration

(C) Tertiary migration

(D) Cross migration

145. The definition : "the pore spaces – connected or disconnected resulting through alteration of a rock, commonly by processes such as dolomitization, dissolution or fracturing" belongs to :

(A) Primary porosity

(B) Permeability

(C) Secondary porosity

(D) Effective porosity

146. The migration of the oil from the reservoir to the trap is known as :

(A) Primary migration

(B) Secondary migration

(C) Tertiary migration

(D) Cross migration

147. The layer containing petroleum oil and gas is :

(A) Above that of water

(B) Below water

(C) Between water and sand

(D) Below sand

148. Boiling point of fuel oil is :

(A) 270°C to 340°C

(B) 270°C to 340°C

(C) Above 500°C

(D) 500°C to 600°C

149. Dead organisms are transformed into petroleum and natural gas in :

(A) Absence of air

(B) Presence of air

(C) Presence of sun light

(D) Absence of sunlight

150. What is called Black Gold ?

(A) Petroleum

(B) Coal

(C) Coal Tar

(D) Natural gas

SPACE FOR ROUGH WORK

SEAL